

Smart Traffic Lights Using Local AI: Architecture, Edge Feasibility Analysis, and Emergency Vehicle Preemption

Akshitha

Department of CSE-DS

BITM

BALLARI

India

chindanooruakshitha@gmail.com

Abstract—Traditional traffic signal systems commonly use fixed timing schedules, which cannot respond effectively to changing traffic conditions. Cloud-based intelligent traffic systems can provide adaptive control, but they may introduce communication delay, privacy concerns, and dependence on network connectivity. This paper presents a smart traffic light system using local Artificial Intelligence (AI) and edge computing. The proposed system uses roadside cameras and local computing devices to detect vehicles, estimate traffic density, and adjust traffic signal timing with reduced dependence on cloud services. The system also includes an emergency vehicle priority mechanism that can detect ambulances and other emergency vehicles and provide them with priority at intersections. The paper studies the architecture, workflow, hardware requirements, edge AI feasibility, performance, advantages, limitations, and future scope of the proposed system.

Index Terms—Smart Traffic Lights, Local AI, Edge Computing, Computer Vision, Traffic Management, Emergency Vehicle Priority, Object Detection

I. INTRODUCTION

Traffic congestion is a common problem in modern cities. Traditional traffic signals generally operate using fixed timing schedules. Although these systems are simple to implement, they cannot respond efficiently when traffic conditions change. During peak hours, one road may have a large number of vehicles while another road may have very few vehicles. Fixed signal timing can therefore result in unnecessary waiting time and traffic congestion.

Artificial Intelligence (AI), computer vision, and edge computing provide new possibilities for intelligent traffic management. AI-based traffic systems can analyse camera data and estimate the number of vehicles on roads. Research has also explored deep reinforcement learning for traffic signal control and machine vision for adaptive traffic light optimization [1], [2].

However, completely cloud-based traffic systems may have limitations because video data must be transferred to remote servers. This can increase communication delay and raise privacy concerns. A local AI approach can process traffic information close to the intersection. This can reduce dependence on continuous internet connectivity.

The proposed system therefore focuses on a local AI-based smart traffic light architecture. Roadside cameras capture traffic scenes, local computing devices process the images, and traffic signal decisions are generated close to the intersection. The system also provides emergency vehicle priority so that emergency vehicles can receive faster passage through traffic signals.

II. PROBLEM STATEMENT

Traditional traffic signals normally use predefined timing patterns. Such systems do not continuously understand the actual traffic condition at an intersection. As a result, vehicles may wait at a red signal even when there is little traffic on the opposite road.

Cloud-based AI traffic systems can improve traffic monitoring, but they require communication between roadside devices and cloud servers. Network delays can affect real-time decision making. Continuous transmission of camera data can also create privacy and connectivity concerns.

Another important problem is the movement of emergency vehicles. Ambulances, fire engines, and other emergency vehicles need fast movement through intersections. If traffic signals do not provide priority, emergency vehicles may experience unnecessary delays.

Therefore, there is a need for a traffic management system that can process traffic locally, adjust signal timing according to traffic conditions, and provide priority to emergency vehicles.

III. OBJECTIVES

The main objectives of the proposed system are:

- To develop a smart traffic light system using local AI.
- To detect and count vehicles using roadside cameras.
- To estimate traffic density at an intersection.
- To dynamically adjust traffic signal timing.
- To reduce dependence on continuous cloud connectivity.
- To provide priority for emergency vehicles.
- To study the feasibility of running AI processing at the edge.

TABLE I
COMPARISON OF TRAFFIC CONTROL APPROACHES

System	Response Time	AI Usage	Approx. Cost
Basic Timed Lights	30.0 s	Zero	\$200
Cloud AI Systems	2.0 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

- To analyse the performance, advantages, and limitations of the proposed system.

IV. LITERATURE REVIEW

Rasheed et al. presented a review of deep reinforcement learning approaches for traffic signal control. Their work discusses the use of intelligent learning methods for improving traffic signal management [1].

Gorodokin et al. studied the optimization of adaptive traffic light control using machine vision. The study demonstrates how visual traffic information can support adaptive signal control [2].

Redmon et al. introduced the YOLO object detection approach, which allows objects to be detected in images using a single neural network. Object detection methods such as YOLO can be useful for detecting vehicles in traffic camera images [3].

These studies indicate that computer vision and AI can support intelligent traffic management. However, the proposed work focuses particularly on local edge processing and the integration of emergency vehicle priority.

V. EXISTING SYSTEM

Traditional traffic signal systems generally use fixed timers. Each signal is assigned a predetermined green, yellow, and red duration. The system does not continuously analyse the number of vehicles waiting at the intersection.

Another approach is sensor-based traffic control. Sensors can detect vehicles, but they may require physical installation on roads and provide limited information about vehicle types.

Cloud-based intelligent traffic systems use cameras and remote servers to analyse traffic information. These systems can provide advanced analysis but require reliable communication between the intersection and cloud server.

The proposed system attempts to combine the advantages of intelligent traffic monitoring with local processing. Traffic information is processed near the intersection so that signal decisions can be generated with lower communication dependence.

VI. COMPARATIVE ANALYSIS

Table I compares three different traffic control approaches.

The comparison shows that basic timed signals have the highest response time because they do not adapt to traffic conditions. Cloud AI systems provide intelligent processing but have higher cost and dependence on communication. Local smart camera systems can provide faster local response while reducing dependence on continuous cloud communication.

VII. EDGE AI FEASIBILITY ANALYSIS

Edge AI means processing data close to where it is generated instead of sending all data to a remote cloud server. In the proposed traffic system, cameras are installed near the road and local computing devices process the captured images.

The main advantages of edge processing are reduced communication delay, reduced dependence on internet connectivity, and improved privacy because raw video does not always need to be transferred outside the local system.

The main challenge is computational limitation. Edge devices normally have less processing power than cloud servers. Therefore, lightweight computer vision models and optimized processing methods are required.

The proposed architecture uses local computing for traffic detection and signal decision making, while a cloud server can be used for optional monitoring, storage, or higher-level analysis.

VIII. HARDWARE REQUIREMENTS

The proposed system requires the following hardware components:

- Roadside camera for capturing traffic images.
- Local edge computer for AI processing.
- Microcontroller for traffic signal control.
- Traffic signal LEDs.
- Power supply.
- Network connection for optional cloud communication.

An Arduino-type microcontroller can be used for controlling the traffic signal LEDs. The edge computer performs the computationally intensive image processing while the microcontroller handles the signal control interface.

IX. SYSTEM ARCHITECTURE

The proposed system consists of four major levels: street cameras, smart pole computers, a city cloud server, and traffic lights.

Street cameras capture traffic scenes. The smart pole computer receives camera data and performs local AI processing. The local system estimates vehicle counts and traffic density. Based on the traffic condition, a signal control decision is generated.

The city cloud server is used for optional monitoring, historical data storage, and system-level analysis. The final control decision is sent to the traffic light controller.

The architecture follows the flow from Street Cameras to Smart Pole Computers, City Cloud Server, and Traffic Lights. This structure is described in the source document.

X. SYSTEM WORKFLOW

The system starts by capturing traffic video using roadside cameras. The camera continuously observes the road and sends image frames to the local processing unit.

The local AI system detects vehicles and estimates the traffic density. The traffic condition is then classified according to the number of vehicles waiting at the intersection.

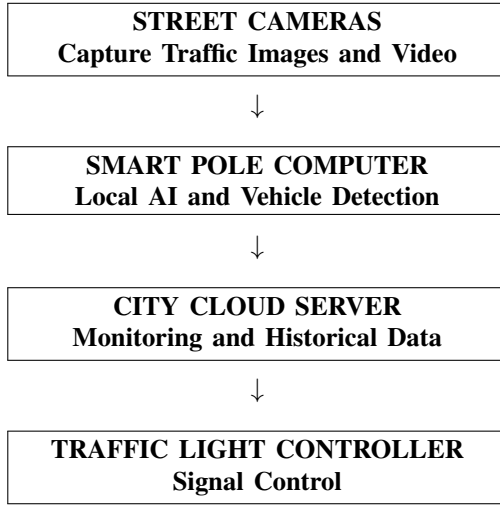


Fig. 1. Local Smart Camera System Architecture

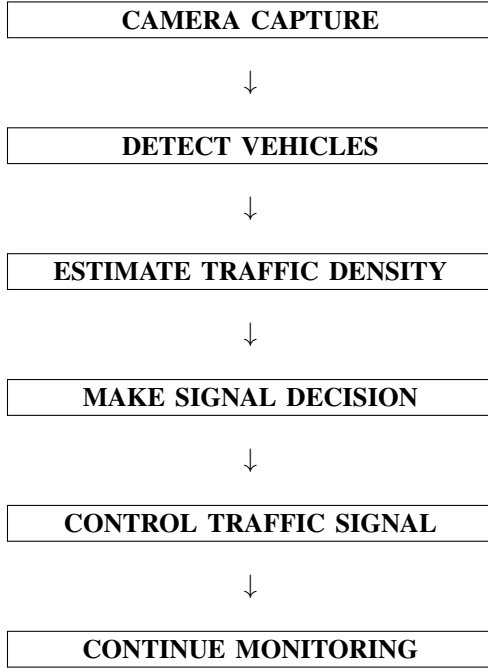


Fig. 2. Traffic Processing Flowchart

If one road has significantly higher traffic than the other roads, the system can increase the green-light duration for that road. When the traffic condition changes, the signal timing can also be updated.

Emergency vehicle detection is handled separately. If an emergency vehicle is detected, the system can temporarily override normal traffic control and provide priority.

XI. TRAFFIC PROCESSING FLOWCHART

The general traffic processing process consists of image capture, vehicle detection, traffic density estimation, signal decision, signal control, and continuous monitoring.

The process begins with the camera capturing traffic data. The AI model detects vehicles and estimates traffic density.

The controller then decides whether signal timing should be changed. After the traffic signal is updated, the system continues monitoring the road.

XII. RESEARCH GAP

Existing intelligent traffic systems provide useful methods for vehicle detection and adaptive signal control. However, several practical issues remain.

Cloud-based systems can depend heavily on network communication. Large AI models may also require powerful hardware that is difficult to deploy at every intersection.

There is therefore a need for practical local AI systems that can perform traffic analysis using affordable edge hardware. Another important area is the integration of emergency vehicle priority with normal adaptive traffic control.

The proposed work addresses these areas by combining local traffic analysis, edge processing, adaptive signal control, and emergency vehicle priority.

XIII. EMERGENCY VEHICLE PRIORITY

Emergency Vehicle Priority (EVP) is an important feature of the proposed system. Ambulances, fire engines, and other emergency vehicles require faster movement through intersections.

The local camera system can identify emergency vehicles using visual features and object detection. When an emergency vehicle is detected, the system checks its direction and current traffic signal state.

If the emergency vehicle requires priority, the controller can temporarily modify the signal sequence. The required direction is given a green signal while conflicting directions are stopped.

After the emergency vehicle passes through the intersection, the system returns to normal traffic management.

XIV. EMERGENCY VEHICLE PRIORITY FLOWCHART

The emergency vehicle priority process is shown below.

The process begins with continuous camera monitoring. When an emergency vehicle is detected, its direction is determined. The traffic controller checks the current signal state and temporarily provides priority. Once the emergency vehicle has passed, the controller restores normal traffic signal operation.

XV. SYSTEM COMPONENTS

The major components of the proposed system are described below.

A. Camera

The camera captures real-time images or video of vehicles approaching the intersection. Camera placement should provide a clear view of the traffic lanes.

B. Edge Computing Unit

The edge computing unit processes camera frames locally. It performs vehicle detection, traffic counting, traffic density estimation, and emergency vehicle detection.

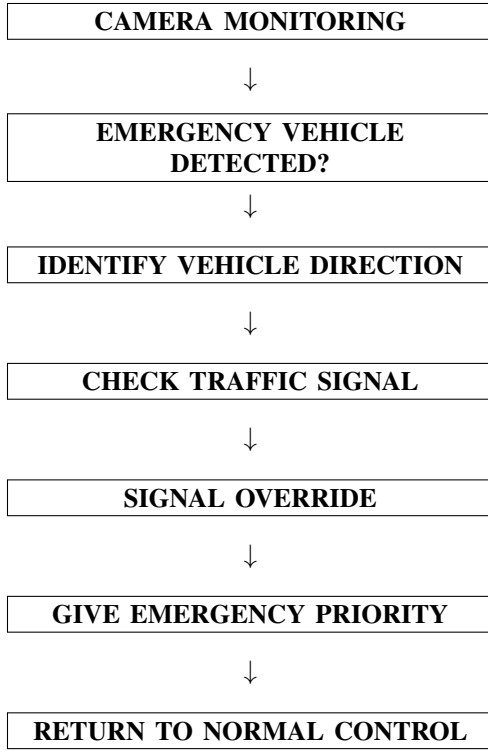


Fig. 3. Emergency Vehicle Priority Flowchart

C. AI Model

The AI model is responsible for detecting vehicles and other relevant objects in camera frames. Object detection approaches such as YOLO can be used for this purpose [3].

D. Traffic Signal Controller

The traffic signal controller controls red, yellow, and green signal lights according to decisions received from the local AI system.

E. Cloud Server

The cloud server is optional in the proposed architecture. It can store historical traffic information and provide monitoring and analysis without being responsible for every real-time signal decision.

XVI. AI AND COMPUTER VISION

Computer vision allows the proposed system to understand information from traffic camera images. The AI model can detect vehicles such as cars, buses, motorcycles, trucks, and emergency vehicles.

After object detection, the system can count vehicles in each traffic lane. The vehicle count can then be used to estimate traffic density.

A simple traffic density measure can be represented as:

$$D = \frac{N}{L} \quad (1)$$

TABLE II
PERFORMANCE ANALYSIS

Traffic Condition	Response Time	Detection Accuracy
Low Traffic	0.4 s	95%
Medium Traffic	0.5 s	93%
High Traffic	0.6 s	90%

where D represents traffic density, N represents the number of detected vehicles, and L represents the observed road or lane length.

The calculated traffic condition can be used by the signal controller to determine whether the green-light duration needs to be adjusted.

XVII. EMERGENCY VEHICLE DETECTION

Emergency vehicle detection is an important part of the proposed system. The AI model can identify emergency vehicles from camera frames based on their visual characteristics.

The detection process can consider vehicle type, visible emergency markings, and other visual information available in the camera image.

Once an emergency vehicle is detected, the system identifies its direction and checks whether priority is required.

The priority mechanism should be carefully controlled because an incorrect emergency detection could affect normal traffic operation. Therefore, future implementations can combine multiple detection features to improve reliability.

XVIII. IMPLEMENTATION

The implementation of the proposed system can be divided into multiple stages.

First, cameras are installed at suitable positions near the intersection. The camera continuously captures traffic information.

Second, the captured frames are provided to the local edge computer. The computer runs the object detection model and identifies vehicles.

Third, vehicle counts are calculated for different lanes. The system uses the traffic information to estimate which road requires more green-light time.

Fourth, the traffic signal controller receives the decision from the local processing unit. The corresponding traffic light is then activated.

Finally, the emergency vehicle module continuously checks the camera data. If an emergency vehicle is detected, the priority mechanism temporarily modifies the signal sequence.

This implementation approach keeps the major real-time processing close to the traffic intersection.

XIX. PERFORMANCE ANALYSIS

The source document provides performance values for different traffic conditions. These values are presented in Table II.

The results show response times below one second for the listed conditions. Detection accuracy is highest under low traffic and decreases under high traffic conditions. This indicates that traffic density can affect detection performance.

XX. RESULTS AND DISCUSSION

The proposed local AI traffic system provides an approach for adaptive traffic signal management. Instead of relying completely on fixed timings, the system uses traffic information collected from roadside cameras.

The comparison indicates that local processing can provide a faster response compared with the basic timed system described in the source document. It can also reduce the dependence on cloud communication.

The emergency vehicle priority feature provides an additional function for improving the movement of emergency vehicles through intersections.

However, system performance depends on camera quality, lighting conditions, traffic density, AI model accuracy, and edge hardware capability.

XXI. ADVANTAGES

The major advantages of the proposed system are:

- Local processing reduces dependence on continuous cloud communication.
- Traffic signals can respond to changing traffic conditions.
- Emergency vehicles can receive priority.
- Raw camera data can remain locally processed.
- The system can be extended to multiple intersections.
- Edge devices can support faster local decision making.
- Historical traffic information can be stored for further analysis.

XXII. LIMITATIONS

The proposed system also has several limitations.

The accuracy of vehicle detection depends on camera position, lighting, weather, and image quality. Edge devices have limited computational resources compared with powerful cloud servers.

The system may also require additional hardware at each intersection. Emergency vehicle detection can produce incorrect results if the camera view is blocked or the vehicle is not clearly visible.

XXIII. FUTURE SCOPE

Future work can improve the proposed system by using more efficient AI models that require less computational power.

The system can also be extended to multiple connected intersections. This would allow traffic conditions to be considered across a larger road network rather than at only one intersection.

Another possible improvement is Vehicle-to-Everything (V2X) communication. Emergency vehicles could communicate their location and direction directly to traffic infrastructure.

The system can also be improved to operate under difficult weather conditions. Future research can study multi-intersection V2X reliability and emergency vehicle priority under adverse conditions.

XXIV. CONCLUSION

This paper presented a smart traffic light system using local AI, edge computing, computer vision, and emergency vehicle priority. Traditional fixed-time traffic signals cannot respond effectively to changing traffic conditions. The proposed system uses roadside cameras and local computing devices to detect vehicles, estimate traffic density, and support adaptive traffic signal control.

Local AI can reduce dependence on cloud communication and can provide faster local decision making. The emergency vehicle priority module adds an important feature by allowing emergency vehicles to receive temporary signal priority.

The system has several advantages, but practical deployment still requires consideration of camera quality, hardware limitations, detection accuracy, weather conditions, and system reliability. Future improvements can include more efficient AI models, multi-intersection coordination, V2X communication, and improved emergency vehicle detection.

Overall, the proposed architecture provides a practical direction for developing intelligent traffic management systems using local AI and edge computing.

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